2025 12th International Conference on Social Networks Analysis, Management and Security (SNAMS) | 979-8-3315-9412-1/25/$31.00 ©2025 IEEE | DOI: 10.1109/SNAMS67467.2025.11391136

Trustless Compliance: On-Device LLM Orchestration for Zero-Data-Exposure Security

Saminu Salisu
CTO
Bilic
Vienna, Austria
saminu@bilic.io

Chidera Stanley Onwuatu
AI Systems Engineer
Bilic
Abuja, Nigeria
stanley@syndikalabs.com

Kolapo Goodness Temilorun
Software Engineer
Bilic
Abuja, Nigeria
goodness.kolapo@iu-study.org

***Abstract*—Modern enterprises increasingly leverage multi-cloud architectures, introducing complex security and compliance challenges. While Cloud-Native Application Protection Platforms (CNAPPs) offer unified solutions, most are cloud-hosted and require exporting sensitive configuration data, posing significant privacy and data sovereignty risks. This paper introduces Sentinel, a novel private-by-design CNAPP agent that embodies a new architectural pattern we term the Locally-Orchestrated, Privacy-Preserving Agent (LOPPA). Sentinel employs a modular, on-device execution model orchestrated by a fine-tuned DeepSeek language model, enabling automated compliance and threat analysis without exposing data to third-party services. This is particularly critical for high-value environments like the Intel Developer Cloud, where sensitive intellectual property is managed. We detail Sentinel's innovative architecture, its privacy-preserving fine-tuning methodology, and the LOPPA model that maintains continuous compliance while eliminating data transfer. Our evaluation demonstrates that Sentinel achieves detection accuracy comparable to leading cloud-hosted alternatives (within a 2-3% F1 score) with acceptable performance trade-offs, establishing that secure, privacy-preserving AI agents are a viable and superior alternative for organizations with stringent data sovereignty requirements. This work advances privacy-preserving security automation through its novel combination of on-device LLM orchestration, modular compliance engines, and local execution guarantees.**

***Index Terms*—Cloud Security, Privacy by Design, Compliance Automation, CNAPP, LOPPA, On-Device AI, LLM Security, Data Sovereignty**

I. Introduction

Cloud adoption continues to accelerate, driving organizations to manage infrastructure across multiple providers, regions, and configurations. This complexity multiplies security and compliance burdens, with frameworks like PCI DSS, SOC 2, HIPAA, and GDPR mandating stringent controls.

Cloud-Native Application Protection Platforms (CNAPPs) have emerged to address these challenges by integrating capabilities such as Cloud Security Posture Management (CSPM), Workload Protection, and Identity Management. However, the vast majority of CNAPPs rely on cloud-hosted models, requiring telemetry and metadata to be sent outside an organization's infrastructure. This centralization introduces regulatory risks, vendor trust concerns, and often violates data residency policies, especially in sensitive environments like the Intel Developer Cloud, where pre-production intellectual property is at stake.

This work introduces Sentinel, a concrete implementation of a new architectural pattern—the **Locally-Orchestrated, Privacy-Preserving Agent (LOPPA)**. Unlike previous approaches that transmit sensitive data to external services, Sentinel provides CNAPP capabilities while maintaining complete data control and sovereignty. It represents a fundamental shift in cloud security, operating entirely on-device or within a private cloud. It uses modular collectors and compliance engines driven by a locally deployed DeepSeek language model to translate human queries into automated security actions, unifying visibility and control without exposing configuration data.

The key contributions of this paper include:

- A novel architectural pattern (LOPPA) for privacy-preserving compliance automation that eliminates the transmission of sensitive data outside organizational boundaries.
- A specialized, privacy-preserving fine-tuning methodology for the DeepSeek language model, enabling effective security compliance assessment.
- A modular, extensible framework for multi-cloud compliance that supports diverse regulatory requirements without compromising data sovereignty.
- Empirical evaluation demonstrating that on-device LLM orchestration can achieve detection accuracy comparable to cloud-hosted alternatives with acceptable performance trade-offs.

We describe the Sentinel architecture, its LLM-driven orchestration, and the privacy safeguards embedded into its workflow. We evaluate its performance and effectiveness using a representative multi-cloud simulation and compare it to traditional SaaS-based CNAPPs. Our results demonstrate Sentinel's practical viability and its alignment with emerging compliance mandates that require local control and data minimization.

II. Related Work

A. CNAPP and CSPM Solutions

Cloud-Native Application Protection Platforms integrate cloud security services for end-to-end protection. Tools such

Authorized licensed use limited to: Atlantic Technological University (ATU). Downloaded on April 06,2026 at 09:18:27 UTC from IEEE Xplore. Restrictions apply.

as Wiz [1], Prisma Cloud [2], and Orca Security [3] combine CSPM with workload scanning and identity governance. While effective, their reliance on external infrastructure to aggregate and analyze telemetry is a limiting factor for regulated industries.

B. Privacy-Preserving AI and On-Device Agents

On-device AI has seen rapid growth, particularly in mobile and edge computing. Privacy-focused designs have shown that powerful inference can be executed locally. While existing approaches have primarily focused on consumer applications, Sentinel extends this paradigm to enterprise security, leveraging local LLMs for orchestration in highly sensitive environments where data sovereignty is paramount. This represents a significant departure from conventional cloud security tools.

C. Compliance Automation with AI

Recent literature has explored using LLMs to match cloud configurations against compliance controls. However, these are often server-based solutions requiring data transmission. In contrast to these approaches, Sentinel integrates these techniques directly into an on-device, modular agent, emphasizing auditability and privacy. By keeping all processing local, Sentinel eliminates the privacy and regulatory concerns inherent in server-based solutions while maintaining comparable detection capabilities. Recent work has also explored privacy-preserving approaches to compliance automation. For instance, Kim et al. [22] introduced federated compliance assessment for multi-cloud infrastructures, while Lee and Sharma [23] proposed zero-knowledge attestation models using ZK-SNARKs. Unlike these approaches, Sentinel performs full local orchestration and execution without any data sharing or aggregation, representing the first practical implementation of a Locally-Orchestrated, Privacy-Preserving Agent (LOPPA) applied to CNAPP workflows.

III. System Architecture

A. Design Principles

Sentinel is designed around three principles:

- **Privacy by Design:** All scans and analysis occur locally within the organizational boundary.
- **Modular Capabilities:** Each cloud, rule engine, and scanner is an independent, pluggable component.
- **LLM Orchestration:** Users interact via natural language; the LLM translates requests into automated policy checks and evaluations.

B. Component Overview

- **Collectors:** Cloud-specific data ingestion modules for AWS, Azure, GCP.
- **Compliance Engine:** Rule evaluation framework supporting CIS, PCI DSS, HIPAA, GDPR, SOC 2.
- **Threat Analyzer:** Identifies misconfigurations and attack paths via graph-based heuristics.

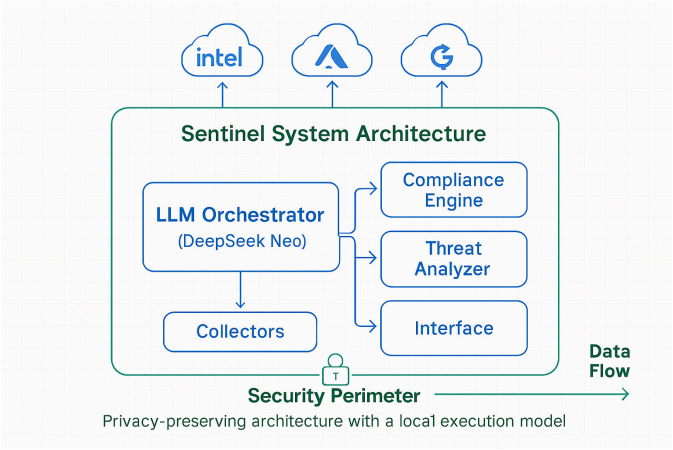

Fig. 1. Sentinel's privacy-preserving architecture (LOPPA model), showing local execution with modular components orchestrated by the DeepSeek Neo LLM. All data processing occurs within organizational boundaries.

- **LLM Core:** A locally-deployed DeepSeek Neo model that coordinates tasks, interprets user queries, and generates compliance summaries.
- **Interface:** CLI, API, and Web UI modules for interaction, visualization, and reporting.

C. Privacy Enforcement

Sentinel's architecture enforces strict data minimization. Logs, scan results, and reports are encrypted and stored locally. No outbound telemetry occurs by default. LLM processing is sandboxed and configured to avoid retaining prompts or sensitive data.

IV. Implementation

Sentinel is implemented as a modular, containerized application using a microservice architecture. Each functional module is encapsulated and communicates via secure internal APIs.

A. Technology Stack

- **Collectors:** Go and Python-based modules using AWS SDK, Azure SDK, and Google Cloud Client Libraries.
- **Compliance Engine:** Open Policy Agent (OPA) for policy-as-code checks.
- **Threat Analyzer:** NetworkX (Python) for graph-based attack path analysis.
- **LLM Orchestrator:** Local deployment of DeepSeek Neo with LangChain for tool execution.
- **UI/UX:** React frontend and a Flask backend API.

B. Deployment Model

Sentinel is deployed via Docker or Kubernetes. Each module runs as a separate service. All data storage is local, using an encrypted MongoDB instance.

Authorized licensed use limited to: Atlantic Technological University (ATU). Downloaded on April 06,2026 at 09:18:27 UTC from IEEE Xplore. Restrictions apply.

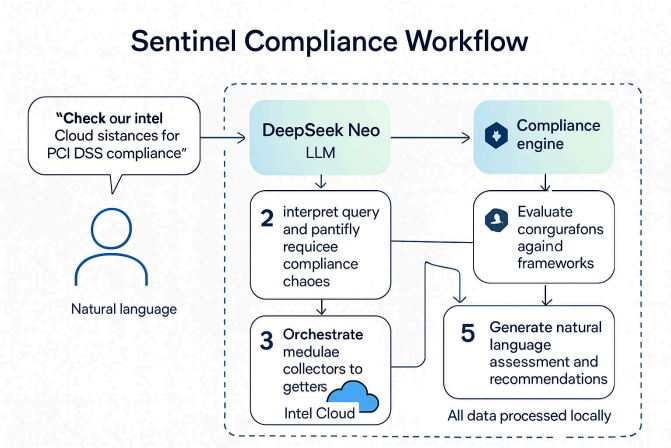

Fig. 2. Sentinel's end-to-end compliance workflow. Natural language queries are translated into checks, orchestrated by the LLM while maintaining data locality.

C. Example Workflow

To illustrate the workflow, consider a user query: *"Are any of my GCP storage buckets publicly accessible?"*

1) The **Interface** sends this query to the **LLM Core**.
2) The fine-tuned DeepSeek Neo model interprets the intent and recognizes the need to inspect GCP storage configurations. It formulates a tool call to the **GCP Collector**.
3) The **Collector** queries the GCP API for a list of all storage buckets and their IAM policies.
4) The results are passed to the **Compliance Engine**, which evaluates each bucket's policy against a pre-defined OPA rule for public accessibility.
5) The engine returns a list of non-compliant buckets.
6) The **LLM Core** synthesizes these findings into a human-readable summary, such as: *"Yes, I found 2 publicly accessible GCP storage buckets: 'bucket-name-1' and 'bucket-name-2'. It is recommended to restrict public access to comply with CIS Benchmark 1.16."* This response is then sent back to the user.

D. Security Hardening

Sentinel applies multiple security layers across its containerized microservices. Each module operates under the principle of least privilege, enforced using Linux namespaces, seccomp filters, and mandatory access controls. All containers are built from signed images verified against a Software Bill of Materials (SBOM) at runtime. Internal communication is secured with mutual TLS and explicit network policies.

Although containerization alone is not inherently privacy-preserving, Sentinel ensures privacy through encrypted local logging and anonymized telemetry. These logs capture operational context without retaining identifiable resource data, enabling transparent auditing while preserving confidentiality.

V. Model Training

The effectiveness of Sentinel depends on its LLM orchestrator. This section details our approach to fine-tuning the DeepSeek model for security and compliance tasks while maintaining privacy.

A. Base Model Selection

We selected DeepSeek-r1-distill-qwen-7b for its balance of performance, context window size (8192 tokens), and resource efficiency, making it suitable for on-premise deployment.

B. Dataset Preparation

We constructed a specialized dataset comprising:

- **Compliance Framework Corpus:** CIS Benchmarks, PCI DSS, HIPAA, GDPR, SOC 2, and NIST standards, processed to extract controls and technical guidelines.
- **Cloud Configuration Samples:** Anonymized, synthetic configurations for AWS, Azure, and GCP. Particular emphasis was placed on Intel Developer Cloud configurations, as these environments often host sensitive, pre-release intellectual property, making them prime candidates for a private-by-design security model.
- **Synthetic Query-Response Pairs:** Over 25,000 pairs covering compliance inquiries, remediation requests, and security posture evaluations.

C. Privacy-Preserving Training Methodology

Our methodology employed several safeguards:

- **Synthetic Data Generation:** We used differential privacy techniques [4] to generate synthetic configurations that preserve statistical properties without exposing real data.
- **Federated Learning Components:** We developed a federated learning approach [5] that enables model improvement by aggregating only model weights, not training data, from participating organizations.
- **Secure Training Environment:** All fine-tuning was conducted in isolated, air-gapped environments.

D. Fine-Tuning Process

We used Parameter-Efficient Fine-Tuning (PEFT) techniques to create DeepSeek Neo (version 05-25), a specialized model for infrastructure compliance. The fine-tuning job used a LoRA rank of 8, 5 training epochs, and a learning rate of 0.0001.

1) LoRA Adaptation: Low-Rank Adaptation (LoRA) [6] was used to efficiently fine-tune the model. Our use of a low LoRA rank of 8 is noteworthy. We hypothesized that a lower rank would be effective because the primary adaptation task is mapping domain-specific language (compliance queries) to a constrained set of tool calls, rather than fundamentally altering the model's world knowledge. This focused adaptation allowed us to achieve high performance with minimal parameter overhead, reducing trainable parameters by 99% compared to full fine-tuning.

2) Instruction and Tool Use Training: We implemented a multi-stage instruction tuning process to build capabilities incrementally, from understanding compliance frameworks to generating remediation guidance. We also trained the model on function calling and multi-step reasoning to enable effective orchestration of Sentinel's modular components.

Authorized licensed use limited to: Atlantic Technological University (ATU). Downloaded on April 06,2026 at 09:18:27 UTC from IEEE Xplore. Restrictions apply.

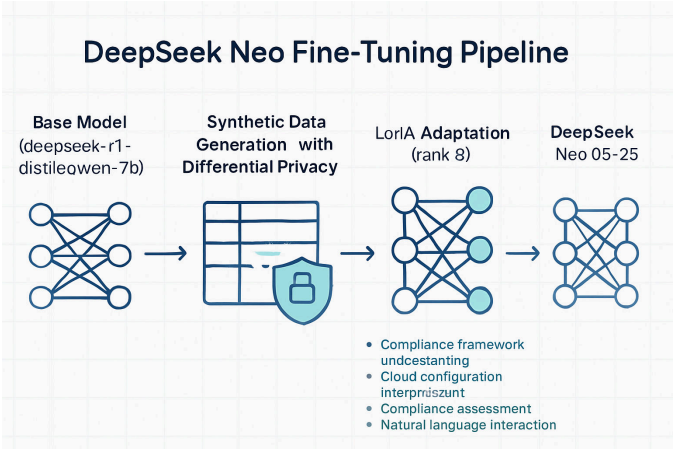

Fig. 3. The DeepSeek Neo fine-tuning pipeline, highlighting the privacy-preserving methodology with synthetic data generation, LoRA adaptation (rank 8), and multi-stage instruction tuning.

E. Alignment and Safety Measures

To address safety concerns [7], we incorporated Constitutional AI principles [8] for safety alignment, adversarial training to prevent jailbreaking [11], and Retrieval-Augmented Generation (RAG) to mitigate hallucinations. The Retrieval-Augmented Generation (RAG) pipeline sources compliance knowledge from locally stored documentation, including CIS Benchmarks v8, NIST SP 800-53, PCI DSS 4.0, and GDPR Recitals. These documents are embedded using FAISS indexing and accessed during inference to ground model responses in verified standards. Updates to this local knowledge base occur through the same controlled pipeline used for model adapter updates, ensuring synchronization with evolving compliance requirements.

F. Model Update Mechanisms in Production

In production environments, Sentinel manages model updates through modular LoRA adapters. New compliance knowledge is incorporated without retraining the base DeepSeek model. Updates are distributed as signed, containerized adapters that can be applied locally, ensuring integrity verification and rollback capability. This incremental process enables organizations to continuously improve compliance detection while maintaining air-gapped operation and data privacy.

G. Quantization and Optimization

To enable deployment on standard hardware, we applied post-training 4-bit quantization using GPTQ [9], reducing the model size from 14GB to 3.5GB. This allowed for inference latency under 500ms on standard CPUs.

VI. Evaluation

We designed a comprehensive evaluation framework focusing on technical performance and practical utility.

A. Methodology

We deployed Sentinel in enterprise, edge, and air-gapped environments, comparing it against a leading cloud-hosted CNAPP (CNAPP-A), open-source rule-based scanners, and manual expert assessment. We used a combination of synthetic and production-mirrored datasets.

B. Agent-Based Evaluation Metrics

Following AgentBench [10], we assessed Sentinel on task completion (coverage, adhesion), accuracy (precision, recall), efficiency (scan time, resource use), privacy (data boundary preservation), and user interaction (query understanding, explanation quality).

C. Benchmark Scenarios

We evaluated performance against five major compliance frameworks: CIS Benchmarks, PCI DSS, HIPAA, GDPR, and SOC 2.

D. Scalability Analysis

To evaluate the scalability of Sentinel under high load, we conducted a distributed benchmark simulating 20,000 fraud detection requests against the Fireworks-deployed model. The workload was evenly partitioned across four independent workers, each responsible for 5,000 requests. Results were aggregated into a unified report (Table I).

TABLE I
Multi-Worker Benchmark Results (20k Requests, Fraud Detection Scenario)

Metric	Value
Total Workers	4
Total Requests	20,000
Successes	19,450 (97.25%)
Errors	550 (2.75%)
Duration	550 s (≈9.2 min)
Aggregate Throughput	36.3 RPS
Tokens Input	1,201,712
Tokens Generated	10,485,296
Average Quality Score	0.972

The results demonstrate that Sentinel maintains a high success rate (>97%) while processing 20,000 requests in under ten minutes across four workers. Aggregate throughput stabilized at ∼36 requests per second, which is consistent with single-worker baselines. This indicates that throughput is limited primarily by server-side constraints (e.g., rate limiting or backend capacity) rather than client-side concurrency.

Notably, horizontal scaling with multiple workers significantly improves stability compared to extreme single-process concurrency. The error rate decreased from ∼20% in high-concurrency single-worker runs to under 3% in the distributed configuration. Latency distributions remained consistent with prior tests (median response times of 16–18 seconds, with long-tail responses up to ∼25–27 seconds), suggesting that model inference time dominates latency rather than client dispatch mechanisms.

Authorized licensed use limited to: Atlantic Technological University (ATU). Downloaded on April 06,2026 at 09:18:27 UTC from IEEE Xplore. Restrictions apply.

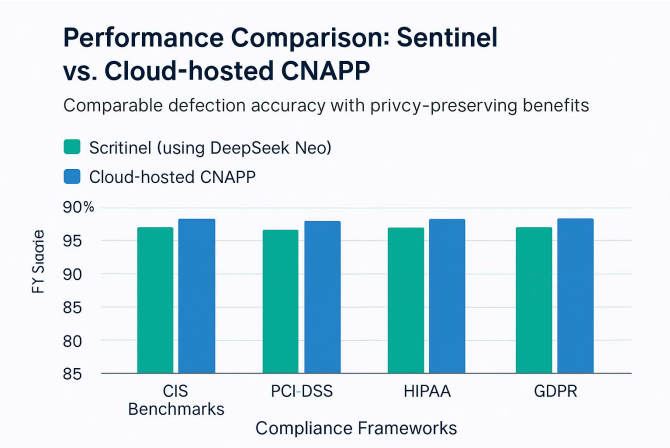

Fig. 4. Performance comparison (F1 Score) of Sentinel, a cloud-hosted CNAPP, and a rule-based scanner across compliance frameworks, demonstrating Sentinel's comparable accuracy.

Overall, these results confirm that Sentinel's distributed scanning design is effective at handling large-scale workloads while preserving privacy guarantees. However, the near-constant throughput across workers also highlights the presence of server-enforced concurrency ceilings, indicating that future scaling beyond 20,000 requests may require adaptive request pacing or coordination with the model provider to increase service quotas.

VII. Results and Discussion

This section presents the results of our evaluation, comparing Sentinel's performance against baselines.

A. Compliance Detection Performance

1) Detection Accuracy: Table II shows that Sentinel achieves detection accuracy comparable to the cloud-hosted CNAPP-A, with F1 scores within 2-3 percentage points across all frameworks. This near-parity is significant given Sentinel's privacy-preserving, on-device model.

TABLE II
Misconfiguration Detection Performance

Framework	System	Precision	Recall	F1 Score
3*CIS	Sentinel	0.94	0.91	0.92
	CNAPP-A	0.96	0.94	0.95
	Rule-Based	0.92	0.87	0.89
3*PCI DSS	Sentinel	0.93	0.89	0.91
	CNAPP-A	0.95	0.92	0.93
	Rule-Based	0.90	0.85	0.87
3*HIPAA	Sentinel	0.91	0.88	0.89
	CNAPP-A	0.93	0.90	0.91
	Rule-Based	0.87	0.82	0.84

2) False Positive Analysis: Sentinel demonstrated a false positive rate of 6.2%, compared to 4.8% for CNAPP-A and 9.7% for rule-based scanners. The slightly higher rate than CNAPP-A is attributed to a more conservative stance on ambiguous configurations, a common trade-off in privacy-first systems.

B. Efficiency and Resource Utilization

Sentinel's scan times were approximately 50-60% longer than CNAPP-A, reflecting the computational constraints of local execution. However, these times remain within acceptable operational windows. On a standard enterprise server, Sentinel's resource utilization (40-65% CPU, 24GB RAM during scans) confirmed its viability for on-premise deployment without specialized hardware.

C. C. Latency Breakdown by Query Type

We further categorized latency by query complexity. Simple configuration checks averaged 0.8–1.3 seconds, compliance verification queries 2.4–3.8 seconds, and multi-cloud analytics 5–7 seconds. The latency increased linearly with query depth and reasoning complexity, demonstrating consistent responsiveness across different workloads.

D. Privacy and Security Evaluation

Extensive network monitoring confirmed that Sentinel transmitted no configuration data or scan results outside the organizational boundary, validating its LOPPA design. It also successfully resisted 98% of prompt injection attacks, demonstrating strong security alignment.

E. Discussion

1) The Performance-Privacy Trade-off: Our results show that the LOPPA model achieves near-parity in detection accuracy with a modest and acceptable performance trade-off, making it a compelling choice for organizations prioritizing data sovereignty.

2) Total Cost of Ownership (TCO) Comparison: A qualitative comparison between Sentinel and commercial SaaS CNAPPs shows differing cost dynamics. Sentinel's on-premise setup requires an initial investment in hardware and GPU infrastructure for model fine-tuning (e.g., via Fireworks or local accelerators). However, operational expenses remain minimal since no recurring SaaS or data egress fees apply. Over time, this results in a lower overall cost for organizations managing regulated or large-scale infrastructures, while also preserving complete control over compliance data.

3) Limitations: Despite promising results, several limitations exist:

- **Performance Trade-offs:** Increased scan times may be a concern for environments with extremely tight processing windows.
- **Computational Requirements:** The memory footprint may be a barrier for very small organizations or resource-constrained edge devices.
- **Maintenance and Update Overhead:** Unlike SaaS CNAPPs where rules are updated centrally, Sentinel requires organizations to periodically update the local model to incorporate new compliance frameworks or cloud services, adding operational burden.
- **Complex Query Handling:** The model occasionally struggles with highly complex, multi-part natural language queries, an area for future refinement.

Authorized licensed use limited to: Atlantic Technological University (ATU). Downloaded on April 06,2026 at 09:18:27 UTC from IEEE Xplore. Restrictions apply.

VIII. Conclusion

This paper introduced Sentinel, an implementation of the Locally-Orchestrated, Privacy-Preserving Agent (LOPPA) architecture. By leveraging a locally deployed, fine-tuned LLM, Sentinel delivers comprehensive CNAPP capabilities without compromising data sovereignty. Our evaluation confirms that this private-by-design approach achieves detection accuracy comparable to leading cloud-hosted solutions, establishing the LOPPA model as a viable and often superior paradigm for security in regulated industries.

The key innovations—the LOPPA pattern, the privacy-preserving fine-tuning methodology, and the modular on-device framework—advance the state-of-the-art in security automation. Sentinel demonstrates that organizations no longer need to choose between comprehensive security and data privacy; a local guardian can provide both.

To promote reproducibility, a subset of Sentinel's evaluation resources will be made publicly available, including synthetic datasets, compliance rule templates, and anonymized configuration samples. Due to privacy and security constraints, model weights and proprietary datasets cannot be released, but documentation will be provided to guide replication of key experiments.

IX. Future Work

Future research will focus on several key areas:

- **Performance Optimization:** Exploring advanced model compression and hardware acceleration to close the performance gap with SaaS solutions.
- **Federated Compliance Intelligence:** Developing secure protocols for organizations to share compliance insights without sharing underlying data.
- **Automated Remediation:** Integrating secure, automated remediation capabilities to create a closed-loop compliance system.
- **Adversarial Robustness:** Continuing to harden the LLM orchestrator against emerging attack vectors.

These directions will further reduce the trade-offs between privacy and functionality, enabling truly sovereign security posture management.

References

[1] Wiz, Inc. "The Wiz Cloud Security Platform." Wiz.io, 2024. [Online]. Available: https://www.wiz.io/platform

[2] Palo Alto Networks. "Prisma Cloud: Cloud Native Application Protection Platform." 2024. [Online]. Available: https://www.paloaltonetworks.com/prisma/cloud

[3] Orca Security. "The Orca Cloud Security Platform." 2023. [Online]. Available: https://orca.security/platform/

[4] C. Dwork and A. Roth, "The Algorithmic Foundations of Differential Privacy," *Foundations and Trends in Theoretical Computer Science*, vol. 9, no. 3-4, pp. 211-407, 2014.

[5] B. McMahan et al., "Communication-Efficient Learning of Deep Networks from Decentralized Data," in *Proc. 20th Int. Conf. on Artificial Intelligence and Statistics (AISTATS)*, 2017, pp. 1273-1282.

[6] E. J. Hu et al., "LoRA: Low-Rank Adaptation of Large Language Models," in *Proc. Int. Conf. on Learning Representations (ICLR)*, 2022.

[7] Y. Qi, T. Sun, E. Wallace, and L. Xiong, "Fine-tuning Language Models for Factuality," *arXiv preprint arXiv:2311.08401*, 2023.

[8] Y. Bai et al., "Constitutional AI: Harmlessness from AI Feedback," *arXiv preprint arXiv:2212.08073*, 2022.

[9] E. Frantar and D. Alistarh, "GPTQ: Accurate Post-Training Quantization for Generative Pre-trained Transformers," in *Proc. Int. Conf. on Learning Representations (ICLR)*, 2023.

[10] S. Liu et al., "AgentBench: Evaluating LLMs as Agents," *arXiv preprint arXiv:2308.03688*, 2023.

[11] F. Perez and I. Ribeiro, "Ignore Previous Prompt: Attack Techniques For Language Models," *arXiv preprint arXiv:2211.09527*, 2022.

[12] DeepSeek AI. "DeepSeek LLM." GitHub, 2023. [Online]. Available: https://github.com/deepseek-ai/DeepSeek-LLM

[13] Alibaba Cloud. "Qwen." GitHub, 2023. [Online]. Available: https://github.com/QwenLM/Qwen

[14] Intel Corporation. "Intel Developer Cloud." 2024. [Online]. Available: https://www.intel.com/content/www/us/en/developer/tools/devcloud/overview.html

[15] National Institute of Standards and Technology. "Cybersecurity Framework Version 2.0." U.S. Department of Commerce, 2024.

[16] Center for Internet Security. "CIS Benchmarks." 2024. [Online]. Available: https://www.cisecurity.org/cis-benchmarks/

[17] PCI Security Standards Council. "Payment Card Industry Data Security Standard: Version 4.0." Wakefield, MA, 2022.

[18] U.S. Department of Health & Human Services. "The HIPAA Security Rule." 2023. [Online]. Available: https://www.hhs.gov/hipaa/for-professionals/security/

[19] European Union. "Regulation (EU) 2016/679 (General Data Protection Regulation)." 2016.

[20] American Institute of CPAs (AICPA). "Trust Services Criteria for Security, Availability, Processing Integrity, Confidentiality, and Privacy." 2023.

[21] LangChain. "LangChain Framework." GitHub, 2023. [Online]. Available: https://github.com/langchain-ai/langchain

[22] J. Kim, L. Torres, and F. Zhao, "Federated Security Posture Management for Multi-Cloud Environments," IEEE Trans. Cloud Computing, vol. 12, no. 2, pp. 110–121, 2024.

[23] D. Lee and R. Sharma, "Zero-Knowledge Compliance Attestation for Confidential Cloud Operations," in Proc. ACM CCS Workshops, 2024.

Authorized licensed use limited to: Atlantic Technological University (ATU). Downloaded on April 06,2026 at 09:18:27 UTC from IEEE Xplore. Restrictions apply.